



Compilation Process: WRF

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Introduction:

Once all the required libraries (such as NetCDF, Jasper, and other dependencies) are installed the Weather Research and Forecasting (WRF) model can be compiled. The compilation process involves (i) downloading the WRF source code, (ii) configuring the build options according to the HPC environment and (iii) Compiling the WRF, and (iv) Verifying the compilation. The following steps describe the standard procedure for compiling WRF.

1. Download WRF Source Code:

The WRF source code is available on the official GitHub repository (<https://github.com/wrf-model/WRF>). It can be downloaded either by cloning the repository using Git or by downloading the compressed source file using wget. Below I am providing the both the procedures.

Using Git:

```
git clone https://github.com/wrf-model/WRF.git
cd WRF
git checkout v4.6.0
```

Using WGET

```
wget https://github.com/wrf-model/WRF/archive/refs/tags/v4.6.0.tar.gz
```

After downloading the compressed file, extract it using the following command:

```
tar -xvf v4.6.0.tar.gz
cd WRF-4.6.0
```

2. Configure WRF:

Important:

WRF contains multiple dynamical cores, but the ARW (Advanced Research WRF) Eulerian Mass core (EM) is the one used in most applications. By setting `WRF_EM_CORE=1`, the compilation process includes the required source files and numerical schemes associated with this core. The next step is to configure the build options. The configuration script checks the system environment and provides several compiler options based on the available compilers and libraries.

Export and Run the configuration script using:

```
export WRF_EM_CORE=1 (you are instructing the WRF build system to compile the  
Eulerian Mass (EM) dynamical core)
```

```
./configure
```

After executing this command, a list of compiler options will appear. Select the options which is suitable for your machine. We are using option 20 in param machine.

3. Compile:

After selecting the appropriate configuration option, the model can be compiled. For real-data simulations, the em_real case is commonly used.

Run the compilation command:

```
./compile em_real >& compile.log
```

This command compiles the model and saves all compilation messages in the compile.log file. which can be used later to check for errors or warnings. Depending on the system performance and number of processors, the compilation process may take some time to complete.

```
grep -i error compile.log
```

4. Verify the Installation:

After the compilation finishes, it is important to verify whether the compilation was successful. This can be done by checking the compilation log file for any errors.

```
grep -i error compile.log
```

```
ls -ls main/*.exe
```

If the compilation is successful, the required executables such as wrf.exe and real.exe will be generated in the main directory of the WRF and is ready to be used for simulations.

References:

✓ WRF - User Guide:

https://www2.mmm.ucar.edu/wrf/users/docs/user_guide_v4/v4.2/WRFUsersGuide_v42.pdf